SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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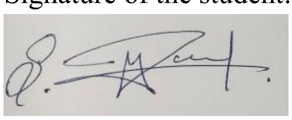
COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct
I **S. Magesh (192525002)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing means using computing resources like servers, storage, applications and processing power through the Internet. Instead of buying and maintaining our own physical servers, we can use resources from cloud providers when needed.

Four characteristics of cloud computing are:

- **On-demand self-service:** Users can get computing resources whenever they need without directly contacting the service provider.
- **Broad network access:** Cloud services can be accessed through the Internet using devices like laptops, mobiles and tablets.
- **Resource pooling:** The provider shares physical resources among many users. The resources are given based on the user's requirement.
- **Rapid elasticity:** Resources can be increased or decreased quickly according to the demand.

In traditional on-premise computing, the organization has to buy hardware, maintain servers and pay for the resources even when they are not fully used. But in cloud computing, resources can be used as needed and payment is usually based on usage.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing developed through many stages of hardware and software evolution.

i. First, mainframe computers were used by many users through terminals. Later, personal computers became popular and computing power was available to individual users.
ii. After that, computer networks and the Internet allowed computers to communicate and share resources. The development of powerful servers and data centers made it possible to provide computing resources from a central location.
iii. Then, virtualization technology was developed. It allowed one physical server to run multiple virtual machines. This improved the utilization of hardware.
iv. Later, the development of web services, APIs and service-oriented software made it possible to access applications and computing resources through the Internet.

These changes finally led to modern cloud platforms. Today, cloud providers use data centers, virtualization, networking and Internet-based software to provide services such as IaaS, PaaS and SaaS.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that divides one physical server into multiple virtual servers. Each virtual server can run its own operating system and applications.

Virtualization is important in cloud computing because it helps to use physical hardware more efficiently. It also allows cloud providers to create and manage virtual machines quickly. Users can get computing resources without directly knowing the physical server.

Virtualization	Cloud Computing
It is a technology used to create virtual machines.	It is a service model that provides computing resources through the Internet.
It mainly focuses on sharing physical hardware.	It provides services like storage, servers and applications to users.
It can be used inside a private organization.	It can provide services through public, private or hybrid clouds.
Example: VMware or Hyper-V.	Example: AWS, Microsoft Azure and Google Cloud.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

SOAP and REST are both used for communication between applications over a network.

SOAP	REST
SOAP is a protocol.	REST is an architectural style.
It mainly uses XML for data exchange.	It can use JSON, XML and other formats.
It uses operations and messages.	It uses HTTP methods like GET, POST, PUT and DELETE.
It is more complex and has strict rules.	It is simple and lightweight.
It is suitable for highly secure and complex enterprise applications.	It is suitable for web and cloud applications.

REST is commonly used in cloud applications because it is simple, lightweight and easy to use with web browsers and mobile applications. SOAP can be used when strong security and strict standards are required.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud service models provide different levels of services to users.

Servic Model	Meaning	Example
IaaS	Provides virtual machines, storage and networking.	Amazon EC2
PaaS	Provides a platform for developing and deploying applications.	Google App Engine
SaaS	Provides ready-to-use software through the Internet.	Gmail
XaaS	Means “Everything as a Service” and provides different services through the cloud.	Microsoft Azure services

- **IaaS:** The user manages the operating system and applications, while the cloud provider manages the physical hardware.
- **PaaS:** The user mainly develops and manages the application, while the provider manages the infrastructure and platform.
- **SaaS:** The complete software is provided to the user through the Internet, so the user does not need to install or maintain it.
- **XaaS:** It is a general model where many different IT services can be provided through the Internet based on the user's requirements.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow